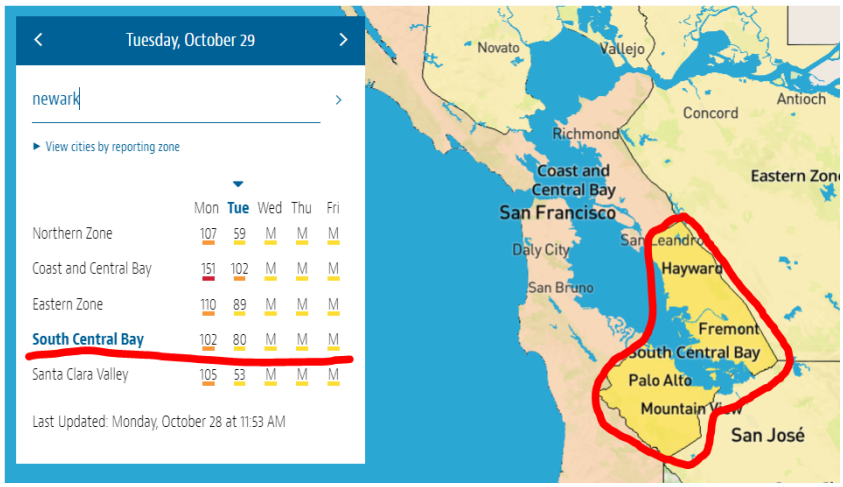
§5141.1 Protection from Wildfire Smoke

Employee Information/Training Document

Oakland—Cal/OSHA is advising employers that steps must be taken to protect workers from harmful exposure if the air quality is unhealthy due to wildfire smoke. California's protection from wildfire smoke standard applies to workplaces where the Air Quality Index (AQI) for fine particles in the air is 151 or greater and where workers may be exposed to wildfire smoke.



When wildfire smoke affects a worksite, employers must monitor the AQI for particulate matter in the air, known as PM2.5. Employers can monitor the AQI using the following:

http://www.sparetheair.org/understanding-air-quality/air-quality-forecast, The City of Newark, is considered to be apart of South Central Bay. If the AQI for PM2.5 is 151 or greater, employers must take the following steps to protect employees:

- Communication – Inform employees of the AQI for PM2.5 and the protective measures available to them.
- Modifications – Implement modifications to the workplace, if feasible, to reduce exposure. Examples include providing enclosed structures or vehicles for employees to work in, where the air is filtered.
- Changes – Implement practicable changes to work procedures or schedules. Examples include changing the location where employees work or reducing the amount of time they work outdoors or exposed to unfiltered outdoor air.
- Respiratory protection – Provide proper respiratory protection equipment, such as disposable respirators, for voluntary use.
    - To filter out fine particles, respirators must be labeled N-95, N-99, N-100, R-95, P-95, P-99, or P-100, and must be labeled as approved by the US National Institute for Occupational Safety and Health (NIOSH).

- If the AQI for PM2.5 exceeds 500, respirator use is required. Employers must ensure that employees use respirators and implement a respiratory protection program as required in California's respiratory standard.

Smoke from wildfires contains chemicals, gases and fine particles that can harm health. The greatest hazard comes from breathing fine particles in the air (called PM2.5), which can reduce lung function, worsen asthma and other existing heart and lung conditions, and cause coughing, wheezing, and difficulty breathing.

Air Quality Index (AQI)	
Categories for PM2.5	*Levels of Health Concern*
0 to 50	Good
51 to 100	Moderate
101 to 150	Unhealthy for Sensitive Groups
151 to 200	Unhealthy
201 to 300	Very Unhealthy
301 to 500	Hazardous

Current Air Quality Index (Current AQI). The method used by the U.S. Environmental Protection Agency (U.S. EPA) to report air quality on a real-time basis. Current AQI is also referred to as the "NowCast," and represents data collected overtime periods of varying length in order to reflect present conditions as accurately as possible. The current AQI is divided into six categories as shown in the table below, adapted from Table 2 of Title 40 Code of Federal Regulations, Part 58, Appendix G.

HOW WE WILL PROTECT YOU:

If employees may be exposed to wildfire smoke, then the employer is required to find out the current AQI applicable to the worksite. If the current AQI for PM2.5 is 151 or more, the employer is required to:

(1) Check the current AQI before and periodically during each shift.

(2) Provide training to employees.

(3) Lower employee exposures.

(4) Provide respirators and encourage their use.

(e) The employer's two-way communication system.

Employers shall alert employees when the air quality is harmful and what protective measures are available to employees. Employers shall encourage employees to inform their employers if they notice the air quality is getting worse, or if they are suffering from any symptoms due to the air quality, without fear of reprisal.

THE EMPLOYER'S COMMUNICATION SYSTEM IS: EMAIL, VOICE CALL, TEXT MESSAGE

(f) The employer's methods to protect employees from wildfire smoke.

Employers shall take action to protect employees from PM2.5 when the current AQI for PM2.5 is 151 or greater. Examples of protective methods include:

(1) Locating work in enclosed structures or vehicles where the air is filtered.

(2) Changing procedures such as moving workers to a place with a lower current AQI for PM2.5.

(3) Reducing work time in areas with unfiltered air.

(4) Increasing rest time and frequency, and providing a rest area with filtered air.

(5) Reducing the physical intensity of the work to help lower the breathing and heart rates.

THE EMPLOYER'S CONTROL SYSTEM AT THIS WORKSITE IS: LIMIT ALL OUTDOOR ACTIVITY (NO PLAYGROUND USAGE) AND REMAIN INDOORS WITH ALL WINDOWS AND DOORS CLOSED WITH A/C ON TO FILER THE AIR. N95 MASK WILL BE MADE AVAILABLE TO STAFF ONLY AND WHO NEED TO MOVE BETWEEN BUILDINGS IF NEEDED.

(g) The importance, limitations, and benefits of using a respirator when exposed to wildfire smoke.

Respirators can be an effective way to protect employee health by reducing exposure to wildfire smoke when they are properly selected and worn. Respirator use can be beneficial even when the AQI for PM2.5 is less than 151, to provide additional protection.
When the current AQI for PM2.5 is 151 or greater, employers shall provide their workers with proper respirators for voluntary use. If the current AQI is greater than 500, respirator use is required. A respirator should be used properly and kept clean.
The following precautions shall be taken:

- Employers shall select respirators certified for protection against the specific air contaminants at the workplace. NIOSH, the National Institute for Occupational Safety and Health of the U.S. Center for Disease Control and Prevention certifies respirators. A label or statement of certification should appear on the respirator or respirator packaging. It will list what the respirator is designed for (particulates, for example).

Surgical masks or items worn over the nose and mouth such as scarves, T-shirts, and bandannas will not provide protection against wildfire smoke. An N95 filtering facepiece respirator, shown in the image below, is the minimum level of protection for wildfire smoke.

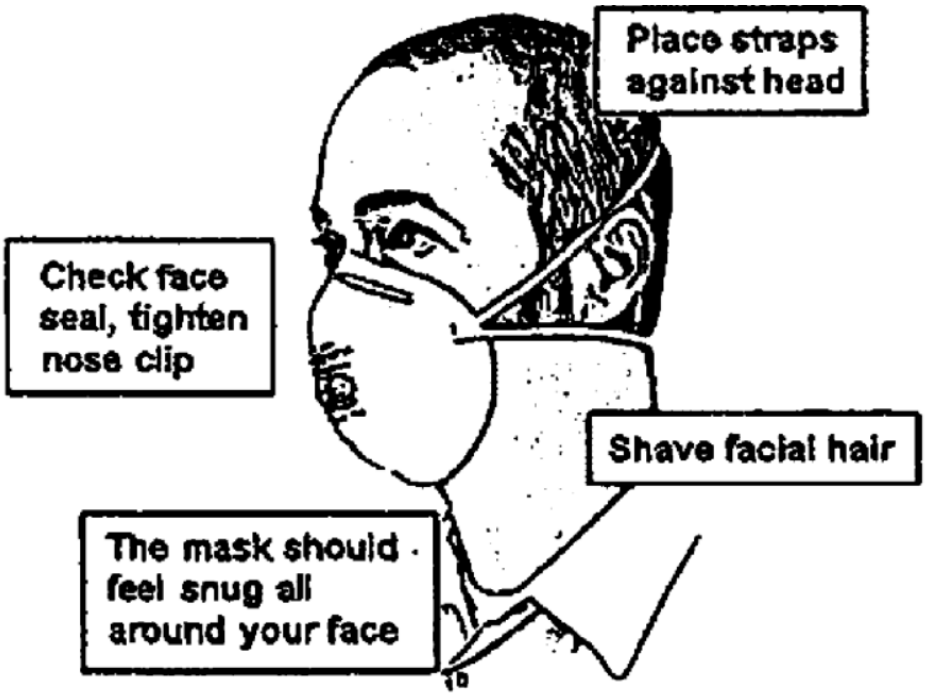
- (2) Read and follow the manufacturer's instructions on the respirator's use, maintenance, cleaning and care, along with any warnings regarding the respirator's limitations. The manufacturer's instructions for medical evaluations, fit testing, and shaving should also be followed, although doing so is not required by Title 8, section 5141.1 for voluntary use of filtering facepiece respirators.
- (3) Do not wear respirators in areas where the air contains contaminants for which the respirator is not designed. A respirator designed to filter particles will not protect employees against gases or vapors, and it will not supply oxygen.
- (4) Employees should keep track of their respirator so that they do not mistakenly use someone else's respirator.
- (5) Employees who have a heart or lung problem should ask their doctor before using a respirator.
- (h) How to properly put on, use, and maintain the respirators provided by the employer.

To get the most protection from a respirator, there must be a tight seal around the face. A respirator will provide much less protection if facial hair interferes with the seal. Loose-fitting powered air purifying respirators may be worn by people with facial hair since they do not have seals that are affected by facial hair.

The proper way to put on a respirator depends on the type and model of the respirator.

For those who use an N95 or other filtering facepiece respirator mask that is made of filter material:

- (1)Place the mask over the nose and under the chin, with one strap placed below the ears and one strap above.
- (2) Pinch the metal part (if there is one) of the respirator over the top of the nose so it fits securely.



Drawing Showing Proper Fitting of a Filtering Facepiece Respirator (shaving is not required for voluntary respirator use)

For a respirator that relies on a tight seal to the face, check how well it seals to the face by following the manufacturer's instructions for user seal checks. Adjust the respirator if air leaks between the seal and the face. The more air leaks under the seal, the less protection the user receives.

Respirator filters should be replaced if they get damaged, deformed, dirty, or difficult to breathe through. Filtering facepiece respirators are disposable respirators that cannot be cleaned or disinfected. A best practice is to replace filtering facepiece respirators at the beginning of each shift.

If you have symptoms such as difficulty breathing, dizziness, or nausea, go to an area with cleaner air, take off the respirator, and get medical help.

https://www.dir.ca.gov/Title8/5141_1b.html

§5141.1 PROTECTION FROM WILDFIRE SMOKE
EMPLOYEE INFORMATION/TRAINING DOCUMENT

Employee Name: _____

Employee Signature: _____

Work Location: _____

Supervisor: _____

Date: _____ Time: _____

I was provided a copy of the following information regarding protection from wildfire smoke, and I'm aware of the information contained in the document and how to protect myself when the AQIs for PM2.5 is 151 or more or is greater than 500, respirator use is required. I have been provided an (N95) mask or similar that meets CAL/OSHA standards and shown how to use it correctly. I will also keep track of my own respirator so that I do not mistakenly use someone else's respirator. If I have any questions, I will reach out to my designated supervisor for clarification and or retraining as needed.

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